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New Generation Sampling and Deep Transient Testing (DTT) Proves Significant Presence of Oil in the Partitioned Zone

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Abstract

A recent exploration well in the northern Partitioned Zone marked a significant milestone after a long pause in drilling activities. Reservoir characterization, focusing on factors like permeability-thickness, skin factor, reservoir boundaries, production potential, and fluid properties, is crucial for assessing reservoir behavior and optimizing production. The well showed high stratigraphic complexity, ranging from tight formations to high permeability zones. Alongside standard and advanced OH wireline logging, new-generation sampling and Deep Transient Testing (DTT) were deployed to understand the above factors. The success of the job demonstrated the value of deploying this new technology for the first time. This abstract highlight the challenges, achievements, and results of the first Sampling-DTT job.

The multi-disciplinary team began planning months ahead to implement the logging program and ensure the necessary equipment was available. While the risk of differential sticking with sampling tools is high due to prolonged pumping, new generation technology has made dynamic well control possible using mud circulation to mitigate this risk. The toolstring featured an innovative focused radial probe with dual flowlines and powerful pumps, reducing sampling and DTT time. This advanced design improved data acquisition efficiency, enabling faster, more precise reservoir evaluations. The multidisciplinary team monitored the job 24/7 for timely decision-making.

Five sampling stations, from Cretaceous reservoirs with one DTT station, were selected based on OH logs and pressure points data. The mobilities range for the stations were ranging from 0.1 mD – 100 mD respectively. All stations showed confidently 100% oil as mobile fluid. The average pumping time in high permeability stations was approximately only 1.5 – 2 hrs and that in tight formation was 3-4 hrs. This was smoothly achievable with high performance pumps in the string. The results were remarkable as well, DTT was able to reach till around 1000 ft inside the reservoir without observing any boundaries and revealing high permeability (Darcy rock) that is laterally extended showing good potential for further development. Also, strong pressure support is observed probably from the water aquifer below the oil zone. The approach of conducting sampling-DTT in exploration well has helped to achieve future cost savings by derisking of reserves uncertainty, determining zonal deliverability and it provides an alternative solution to conventional DST.

Deep Transient Testing (DTT) is an advanced wireline technique that explores higher permeability and thicker reservoir zones with a larger radius of investigation, flowing faster and longer with high-resolution packages, compared to traditional Interval Pressure Transient Testing (IPTT). DTT is not a replacement for Drill Stem Testing (DST), rather bridges the gap between IPTT and DST, especially in multi-reservoir fields where testing all zones is costly. Additionally, DTT supports efforts to reduce carbon dioxide emissions.

Introduction

The Partitioned Zone (PZ), located between the State of Kuwait and the Kingdom of Saudi Arabia, contains several prolific clastic and carbonate reservoirs. Exploration efforts were recently revived in 2024 to evaluate the hydrocarbon potential of mainly the Cretaceous Clastic reservoirs. Fig 1 shows the stratigraphic column of Kuwait highlighting various clastic and carbonate reservoirs. These formations exhibit complex stratigraphy, lateral heterogeneity, and varying mobility, demanding a thorough reservoir evaluation strategy.

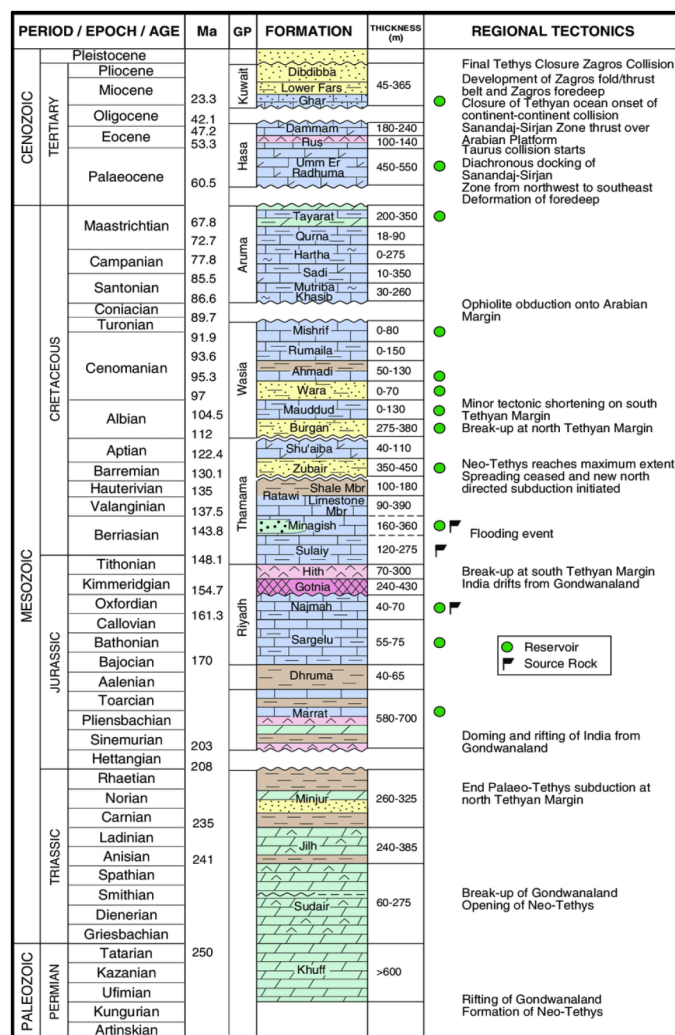


Figure 1—Stratigraphic column of Kuwait

Given the absence of modern dynamic data in this area, comprehensive open-hole logging was the first step in characterizing reservoir quality and fluid saturation. A suite of basic and advanced services – including resistivity imaging, nuclear magnetic resonance, dielectric logs, and formation pressure measurements was deployed to assess multi-physics formation evaluation.

Following the log interpretation, wireline formation fluid sampling was conducted across multiple intervals in both clastic and carbonate zones. This helped to identify fluid mobility, identify contacts, and capture representative samples for PVT analysis. All stations confirmed the presence of mobile oil with varying mobility across the formations.

Subsequently, a Deep Transient Testing (DTT) was designed and executed in the most promising and the main intervals within the clastic reservoir. The DTT station was selected based on a combination of log-derived properties, pressure gradients, and fluid mobility observed during sampling. Unlike conventional short-duration pressure testing, DTT enables multi-hour drawdown and buildup measurements, significantly extending the radius of investigation. This allows for accurate determination of reservoir-scale permeability, pressure support, boundary conditions, and flow regimes.

The objective of this paper is to present the successful and safe job execution of new generation formation testing platform in a clastic reservoir of the exploration field. It highlights the critical role of DTT in reducing reservoir uncertainty, confirming hydrocarbon deliverability, and enabling informed early-stage development decisions. Additionally, the paper aims to showcase the capabilities of the new-generation sampling and testing toolstring in achieving high-quality results within a significant shorter time frame, while ensuring operations safety through dynamic well control and real-time decision making.

New Generation Formation Tester

Wireline formation testing has progressed significantly since its inception in 1950s, evolving from basic pressure measurements to advance pressure gradient profiling, fluid contact identification and dynamic testing capabilities. Modern platforms integrate 3D radial probes, focused sampling, and downhole fluid analysis, enabling acquisition of high-purity formation fluids across diverse reservoir types. The latest generation of tools rated up to 200 °C and 35,000 psi – feature dual flowline systems (0.25-inch flowlines), smart automation, and are designed to operate reliably even in tight, unconsolidated formations with complex fluid systems, setting a new standard for reservoir characterization. The tool is a compact 4.5-inch OD reducing tool weight by 30 % and length by 25 % compared to legacy designs. The details of the new generation formation tester were presented by [Partouche et al. 2020](#).

The downhole hardware of the new generation formation testing platform is composed of modular subsystems, including a focused radial probe, dual packer, fluid analyzer, high-performance pumps, and sample capture units. The tool's electrical capability increases significantly downhole power availability, enabling flow rates of up to 120 cm/s (can be upgraded to 200 cc/s) and allowing autonomous control of key operations such as packer inflation, fluid cleanup and bottle filling.

Focused Radial Probe

The focused radial probe as shown in [Fig. 2](#) has two inlets – guard and sample line inlets radially focused on the tool as shown in the figure. The total flow area of 108 sq.in enables focused sampling thereby reducing drastically the stationary time per station. The differential pressure rating of the packer is 8000 psi. It also has a auto-deflate valve to ensure a fail-safe retraction of the elements in case the tool permanently loses power downhole while it is set. By default, all guard drains are connected to secondary flowline and all the sample drains are connected to the primary flowline. The seal values in the probe can isolate the flowline from each of drains from the rest of the flowline which minimizes the storage volume during a pressure buildup. There are bypass lines which run along the mandrel element for the wellbore pressure equalization in presence of losses below the element.

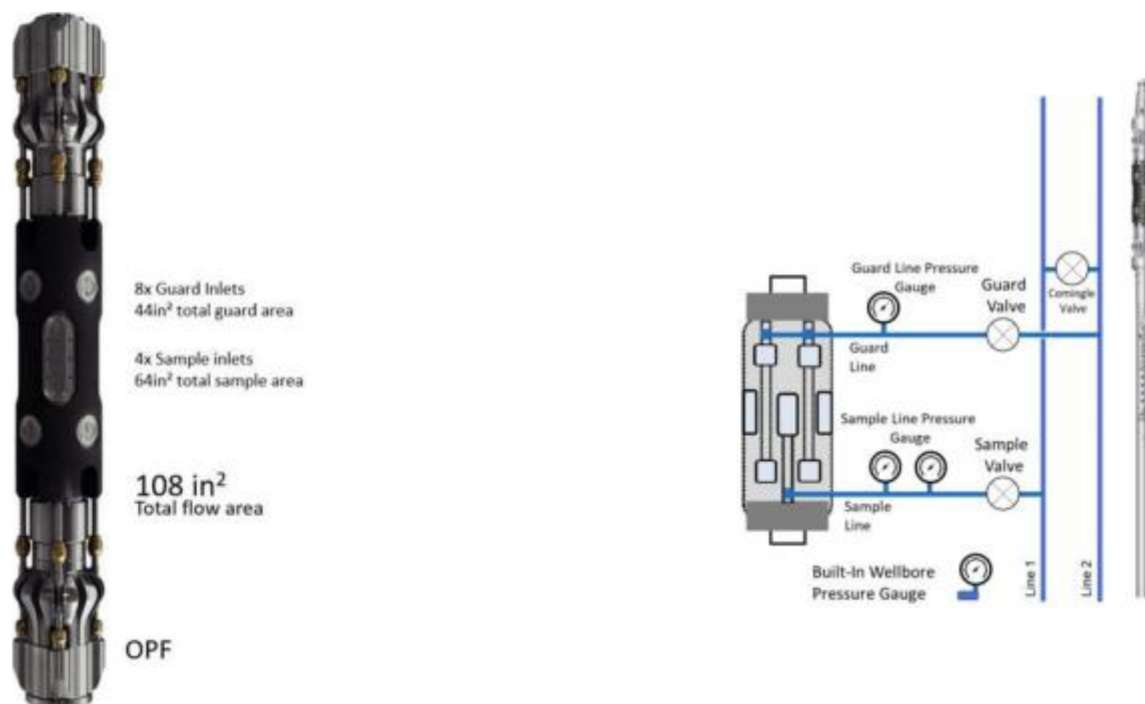


Figure 2—Radially focused probe & simplified flowline of the probe

Pump Module

The sampling and DTT toolstring consisted of dual pumps (5K and 8K differential). Each pump delivers 0.2 – 60 cc/s for 5K differential unit and 0.1 – 40 cc/s for 8K differential unit. Having one pump per flow line allows precise focused sampling and doubles flow capacity. For high-mobility reservoirs as described in this paper for DTT operations, the pumps were upgraded to achieve total flow rates up to 200 cc/s, enhancing operational flexibility during both sampling and DTT.

Fluid Analysis Module

The fluid analysis module utilizes a dual-window, 24-channel optical filter spectrometer that simultaneously monitors optical density across both flowlines. Both on sample and guard line there are density, viscosity and resistivity sensors for fluid identification and clean-up monitoring. Resistivity sensor operates on an induction-based design without electrodes to prevent fouling. A fluorescence sensor is present in the flowline. The spectrometer-based analysis enables calculation of hydrocarbon weight fractions (C1 through C6+ and CO₂), formation volume factor (FVF) and gas-oil ratio (GOR).

Sample Carrier Module

The sample carrier module holds six bottles and is fully automated – with motor-driven, instrumented valves capable of multiple cycles. Its design isolates buffer fluid via seal valves that flush dead-volume before sampling, ensuring the bottle headspace and elastomers are saturated – critical for accurate trace-element measurements (e.g., H₂S). The bottles self-seal to prevent sample loss at the surface. Hydrostatically compensated versions of 675 cc were used in the job described in this paper for sampling purposes.

Pressure Gauge

The new formation testing tool incorporates a new gauge design with higher resolution, which leads to improved signal-to-noise ratio. For pressure transient analysis, one important pressure gauge characteristic is gauge resolution, which is the minimum actual pressure change detected by the gauge. High resolution

quartz gauge in this tool has been set at 0.003 psi at temperatures less than 70 °C and 0.005 psi at temperature higher than 70 °C.

Pressure Transient Analysis

Reservoir testing plays a key role in evaluating flow potential and confirming the presence of hydrocarbons. The main objective of flow tests is to obtain permeability, well deliverability and to confirm reservoir extent. Pressure transient analysis can be applied at different scales, ranging from small pretest and IPTT to full-scale well tests such as DST as shown in Fig. 3 (Ling *et al* 2025). In regions such as the Middle East, reservoirs are often thinly bedded, laminated, and compartmentalized. Conducting a Drill Stem Test (DST) on each target zone in such settings is usually not practical due to high operational costs and limited rig time. Moreover, DSTs would require surface flow and flaring, which increases cost and environment footprint.

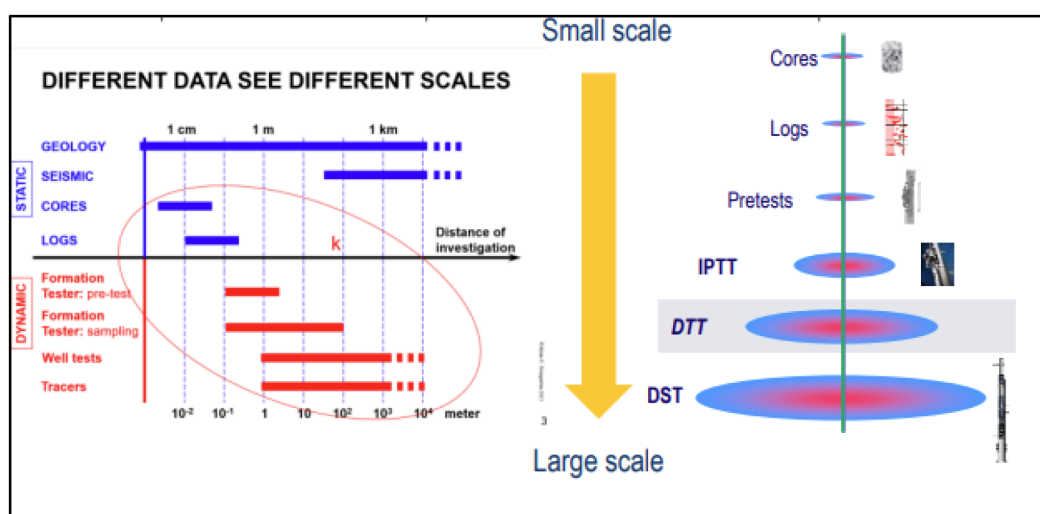


Figure 3—Comparison of investigation scales across reservoir evaluation techniques

As a more cost-effective and environmentally friendly alternative, wireline-conveyed testing methods such as **Interval Pressure Transient Testing (IPTT)** and **Deep Transient Testing (DTT)** have been introduced. These methods allow testing of individual zones without flowing hydrocarbons to surface or requiring extensive surface equipment. Each interval can be isolated and evaluated, making them ideal for stacked or multi-layered reservoirs.

IPTT has been widely used for over two decades to estimate zonal permeability. It is effective for testing single, hydraulically connected layers, and the results often correlate well with those from DST. However, IPTT has limitations in reservoirs with low permeability, high fluid viscosity, or thick pay zones, where longer test durations and larger fluid volumes are needed. The DTT concept brings distinct advantages over IPTT in terms of (1) flow rate, (2) flow duration and (3) pressure gauge resolution.

The introduction of **DTT technology**, especially with the use of a circulating sub, has overcome many of these limitations. DTT enables extended flow periods and larger fluid volumes to be pumped safely without compromising wellbore stability. The improved design allows for higher flow rates and better pressure resolution, resulting in a much deeper radius of investigation. As a result, DTT has made it possible to evaluate even high-permeability or gas-bearing reservoirs more accurately and efficiently.

Deep Transient Testing (DTT) – A Breakthrough with New Generation Tool

DTT is a modern formation technique designed to reach deeper into reservoir than conventional IPTTm making it suitable for high-permeability, thick formations, and low-viscosity fluids. It enables real-time

measurements mainly of pressure, temperature, viscosity, density, GOR and fluid composition. During a DTT, formation fluid is pumped at high rates (up to 200 cm³/sec or ~108 bbl/day), creating a significant pressure drawdown. When pumping stops, pressure builds up as fluid from the reservoir refills the region, allowing accurate estimation of formation permeability, anisotropy, and other properties.

This new generation tool has advanced technology components, including powerful pumps and improved gauges, ensure longer and clearer pressure signals, with extended buildup periods reaching far beyond near-wellbore damage. Although prolonged high-rate pumping displaces large hydrocarbon volumes, the use of circulating subs helps maintain well control and avoid underbalance. DTT can deliver data comparable to a drill stem test (DST) while minimizing fluid flow, surface equipment, and emissions. As illustrated in Fig. 4 comparison of DTT vs DST (Kassim *et al.* 2023).

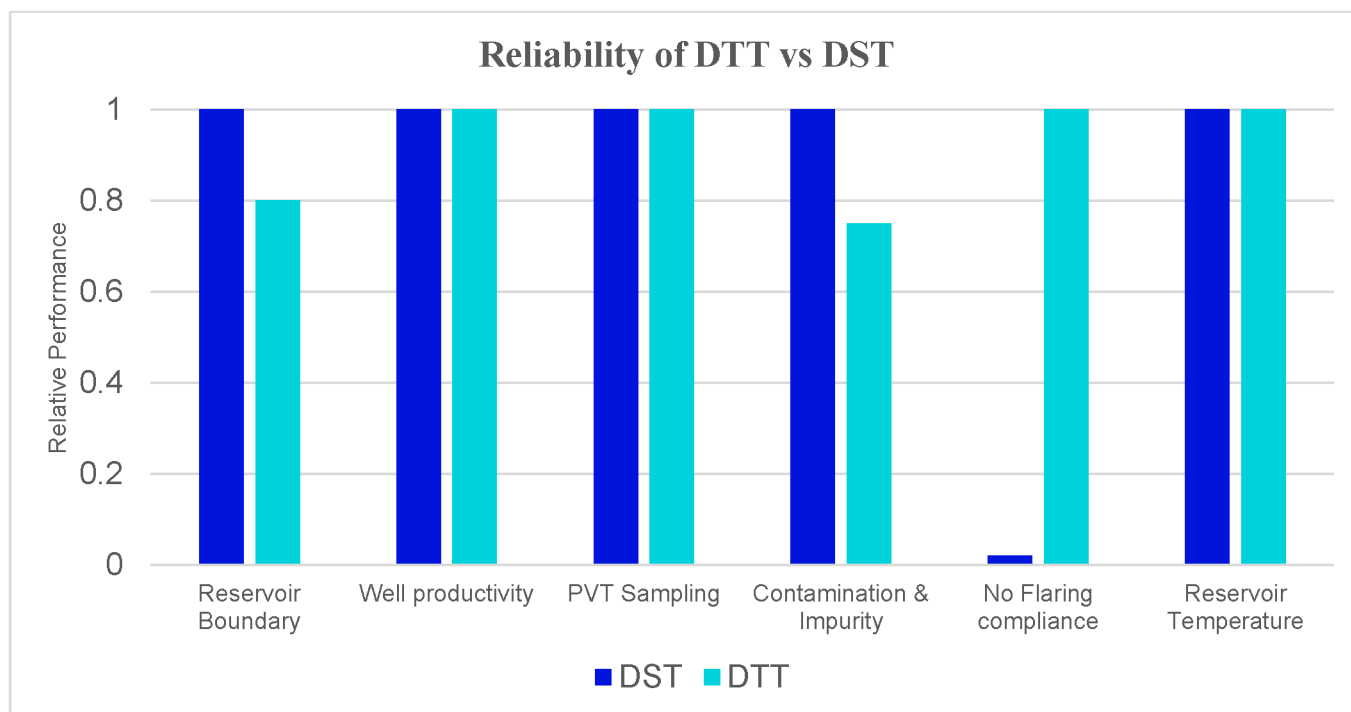
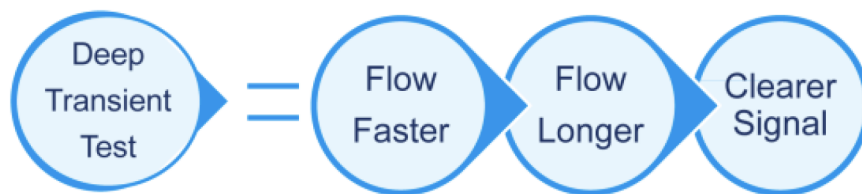


Figure 4—Comparison of DTT vs DST



DTT operations are conducted using a drill pipe-conveyed formation tester, where the tool is deployed below a circulating sub and a slip joint assembly. The circulating sub plays a vital role by enabling the formation fluid pumped from the reservoir to be diluted with circulating mud and safely transported to the surface (Fig. 5).

Circulation is maintained through drill pipe annulus at typical rates between 100 – 250 gal/min, which is approximately 50 to 100 times greater than the actual downhole pump rate. During circulation, the blowout preventer (BOP) annular is closed, and the mud-hydrocarbon mixture is diverted through the choke line to the mud-gas separator.



Figure 5—Circulating sub

Continuous circulation of mud during the flow period can cause thermally induced drill pipe shrinkage. The slip joint serves an important mechanical function by compensating for thermal expansion or contraction of the drill pipe during extended pumping and circulation. As the formation tester (FT) remains anchored at the borehole wall, any pipe movement due to temperature changes could transfer mechanical stress to the tool. The slip joint prevents this by allowing axial flexibility, ensuring the integrity and stability of the formation tester during the test. Finally, the formation tester is equipped with high resolution gauges, which enables longer buildup times before the signal-to-noise limits are reached. A typical DTT string is shown in Fig. 6.

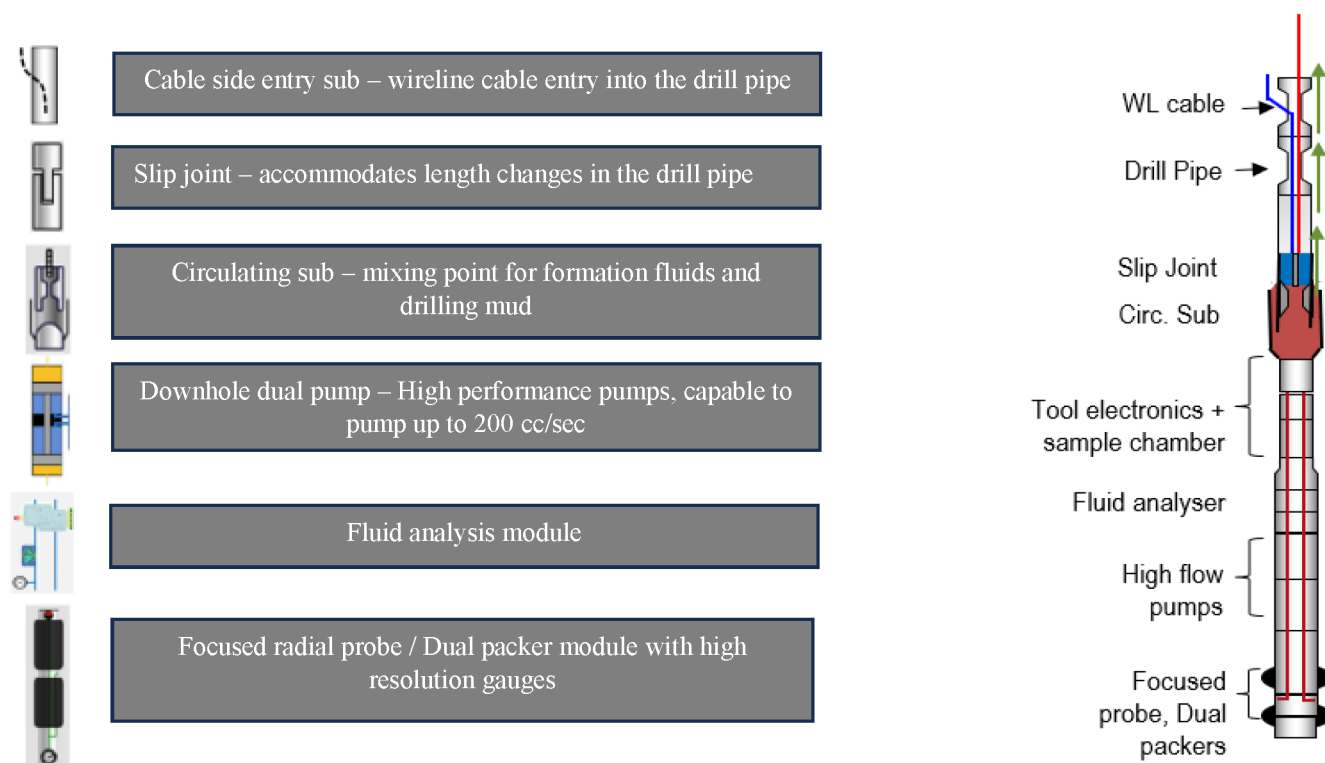


Figure 6—Typical DTT string run downhole

The following are the key components/modules used in DTT operations, along with their respective functions (S.N. Cheng *et al.*, 2025).

Workflow

Exploration well (X-1) was drilled with an 8.5-inch bit, resulting in approximately 1,150 ft of open hole (OH). The OH section targeted the cretaceous reservoir interval and was drilled with water-based mud (WBM). A comprehensive formation-evaluation program was executed to reduce petrophysical uncertainties as per methodology described by [Datir *et. al* 2021](#). Integrated interpretation of multi-physics log measurements from the triple-combo suite, high definition elemental spectroscopy, dielectric dispersion and nuclear magnetic resonance was able to establish important petrophysical parameters across the logged interval.

Elemental spectroscopy measurements ([Craddock *et. al* 2021](#)) in the drilled clastic sequences identified the detailed mineralogical variations and helped to differentiate quartz-rich feldspatic sands from mixed-clay rich intervals. Matrix parameters and Organic carbon measurements from the spectroscopy tool provided additional inputs for determining oil saturation. The data was key in determining the litho-facies variations in the uranium rich carbonates also.

Dielectric dispersion and Chlorine measurements were jointly inverted to provide in-situ water salinity ([Miles *et. al* 2020](#)) and rock textural parameters. The data was used to enhance resistivity-derived water saturation (S_w) in low-resistivity pays. This was specifically required for the texturally rich carbonate sequences and argillaceous clastics where high irreducible water content and pore type variations mask hydrocarbon bearing pays.

Nuclear magnetic resonance and Mineralogy ([Kaushik *et.al* 2021](#)) data was used to determine realistic permeability estimates and irreducible fluid volumes.

The advanced formation evaluation provided the framework for net-pay identification, clearly distinguishing moveable-oil-zones from tight, bound-fluid layers and take informed decisions regarding intervals for dynamic testing. The quick avaluation enabled precise selection of formation-pressure points, five-fluid sampling stations, and the final deep Transient Test (DTT) interval in the most permeable Cretaceous sands. The workflow is illustrated below in the graphical representation as shown in [Fig. 7](#).

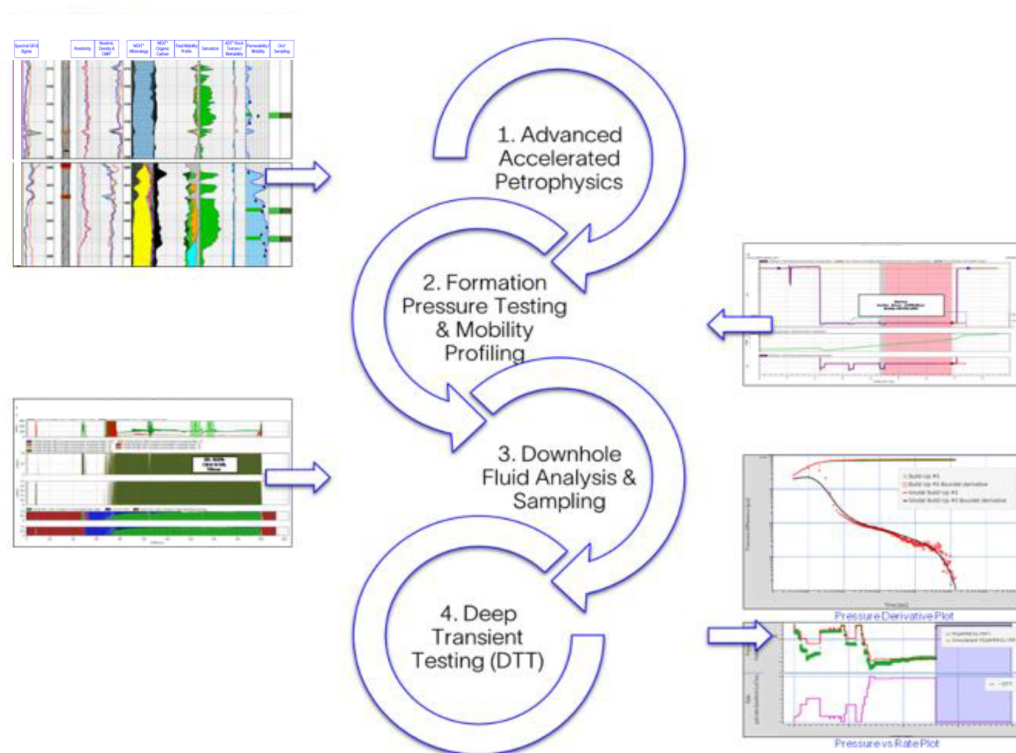


Figure 7—Graphical representation for the workflow conducted for X-1 well

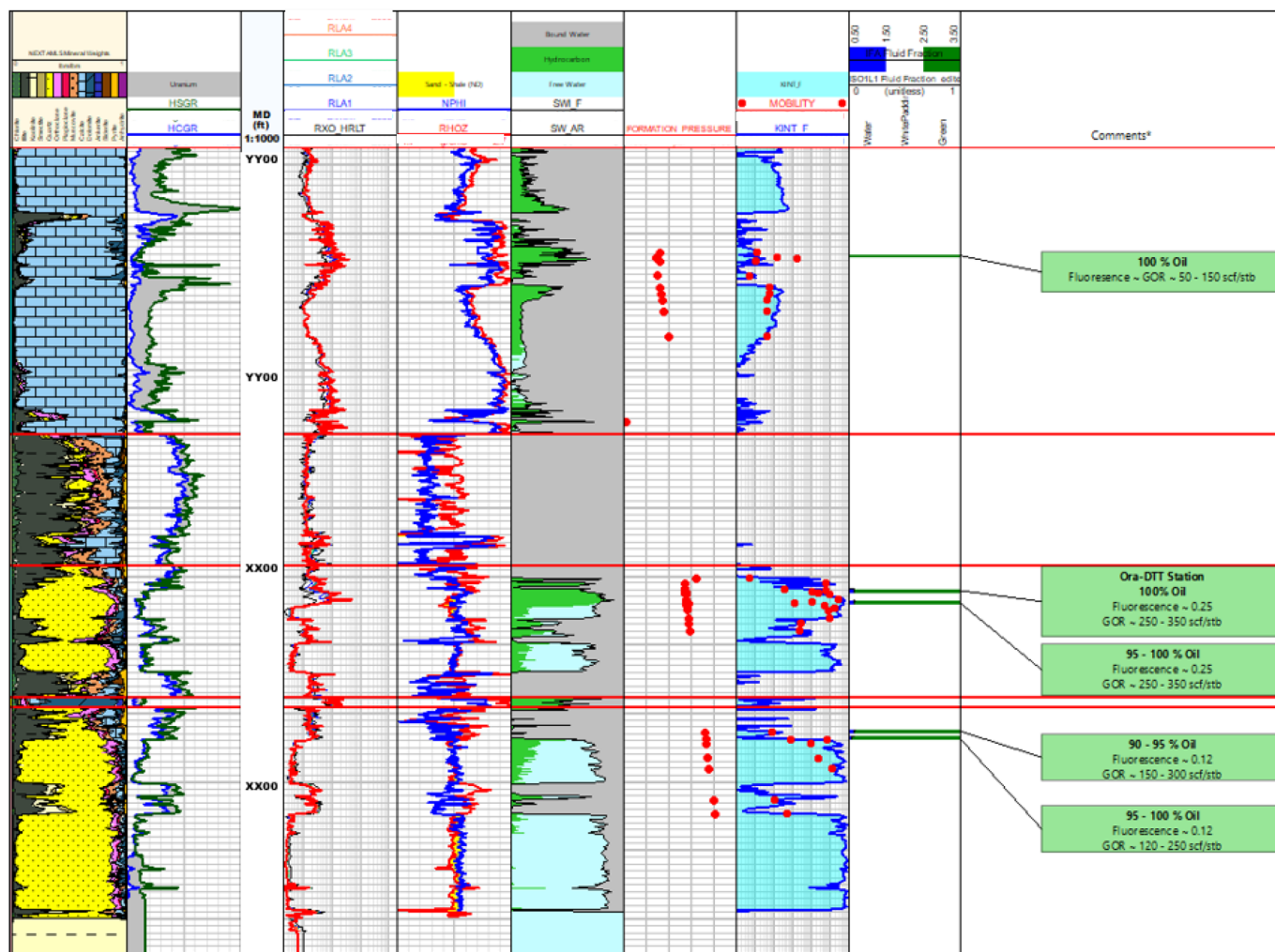


Figure 9—Composite log showing pressure points, mobility & sampling depths

Sampling & Deep Transient Testing

Five fluid-sampling stations were conducted across the Cretaceous interval followed by a DTT station as shown in Fig. 9. Four stations targeted clastic sands, while one addressed the overlying carbonate, mobility ranging from 0.1 – 100 mD/cP. Real-time optical analysis confirmed 100% mobile oil at every station. High-mobility intervals required only 1.5-2 hrs of pumping, whereas in tight clastic formation sampling was completed in 3-4 hrs. This would not have been possible without the high performance pumps.

All four clastic intervals were carried out with the focused radial probe. However, the tighter station at carbonate interval was carried out with the dual packer. Incorporating of dual packer within the new generation formation tester was instrumental in facilitating sampling at tight carbonate reservoir. New generation focused radial probe was critical in expediting the clean-out process, significantly reducing stationary time at sampling stations.

At the designated DTT depth of XX24 ft MD, sampling was completed and both pumps were ramped to a combined rate of ~ 100 cc/s. A total of ~ 1350 L of formation fluid was pumped in ~ 4 hrs, with a controlled drawdown sufficient from transient analysis. The pressure drawdown during the flow period reached approximately 100 psi, which is sufficient to ensure a high signal-to-noise ration by producing strong pressure response that remains clearly distinguishable above the instrumentation and operational noise. The subsequent pressure buildup was taken for another ~ 3.8 hrs provided a clear derivative response

to determine the reservoir permeability and radius of investigation (ROI). Below is the figure 10 showing the pressure, rate & total volume pumped at XX24 ft MD.

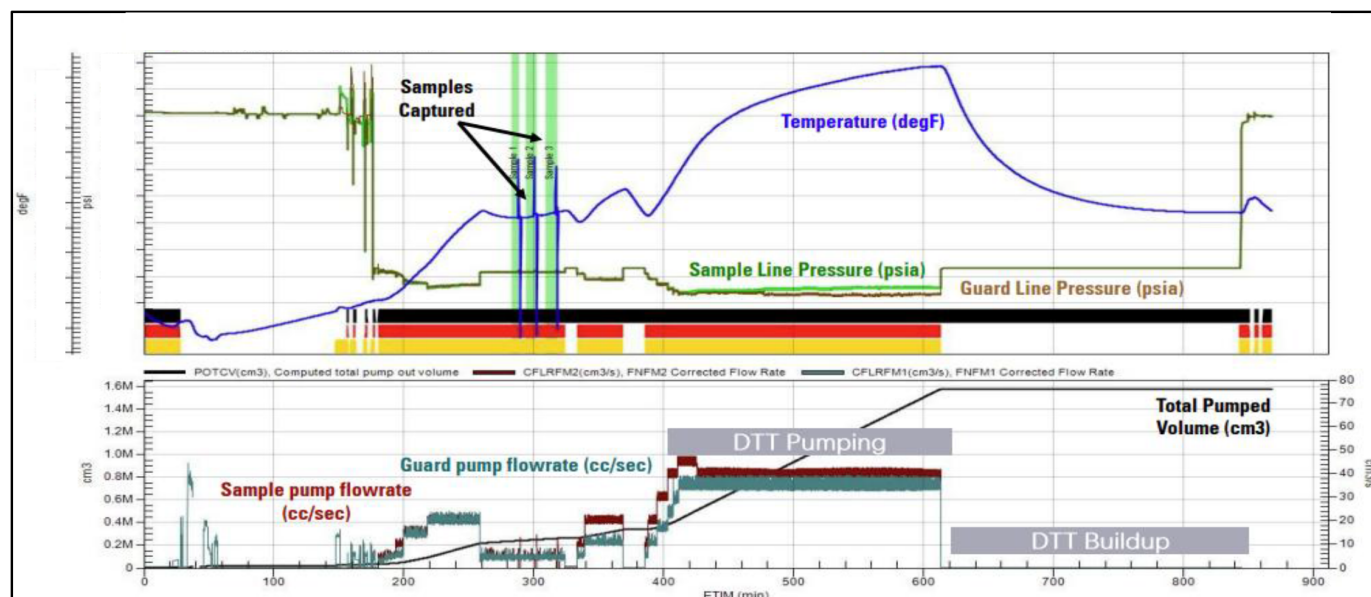
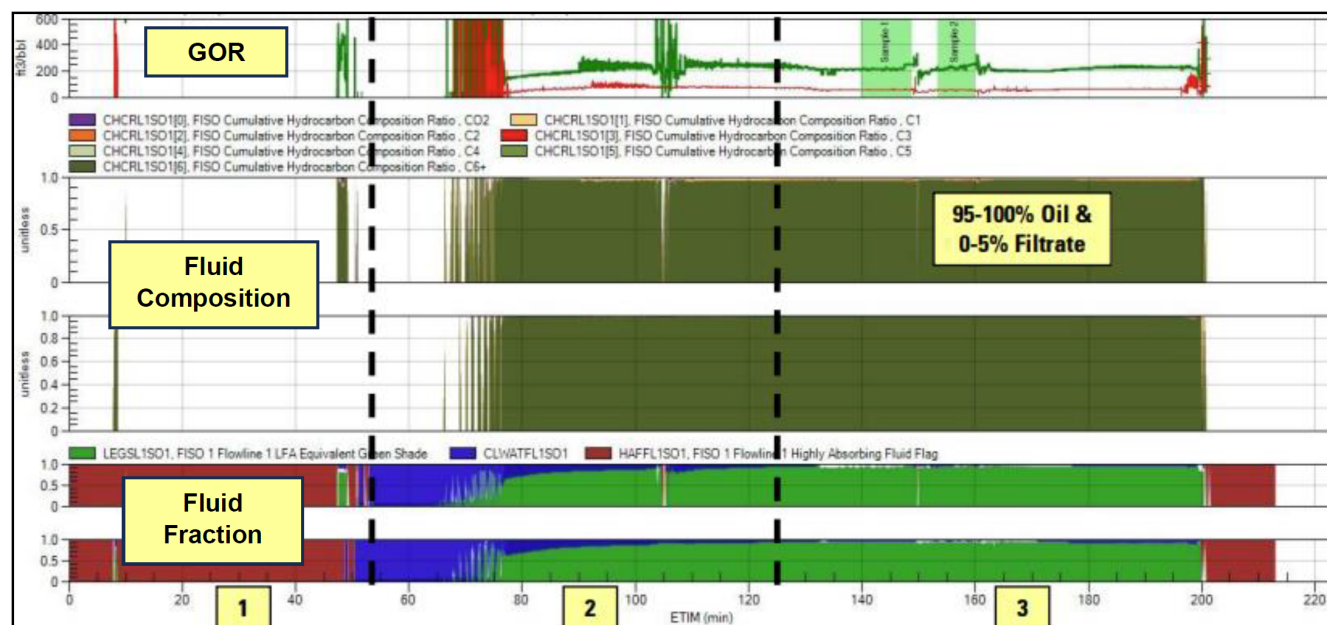


Figure 10—Pressure, Rate, Volume vs Time

Fluid Analyzer as shown in fig. 11, showed good quality of oil (API ~ 24 – 28) is estimated to be flowing across this zone with GOR ~ 250 – 350 scf/stb and fluorescence of ~ 0.24. Fluid composition showed ~ 90 % of C_{6+} and ~ 8 % of the lighter fractions with ~ 2 % of CO_2 .



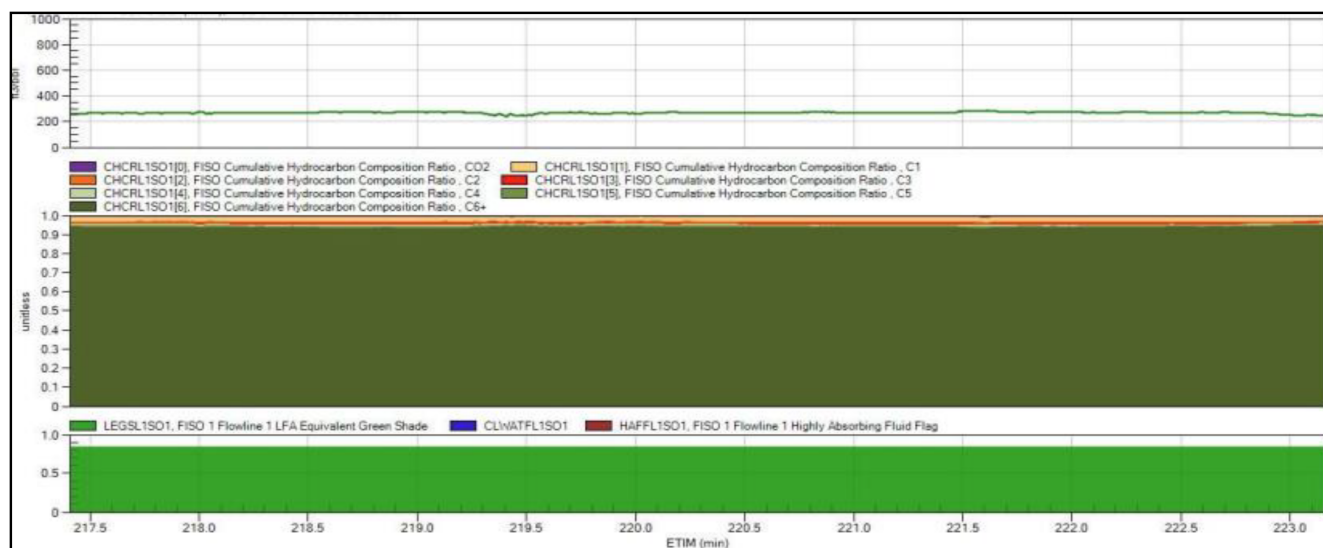


Figure 11—Fluid Analyzer – GOR, fluid fraction & composition

Deep Transient Testing (DTT) Analysis

Focused radial probe was utilized at DTT station. The pressure buildup data during the main buildup was analyzed. Analytical model was built using the buildup data. Interpretation was carried out using KAPPA™ software. Formation parameters such as permeability and radius of investigation were estimated.

The most satisfactory match was obtained by using single layer model. The selection of single layer model was based on OH logs. The [fig 12](#) below illustrates that the full 40 ft interval, as interpreted from open-hole logs, appears productive with no detectable barriers – thus the entire thickness is considered to be contributing.

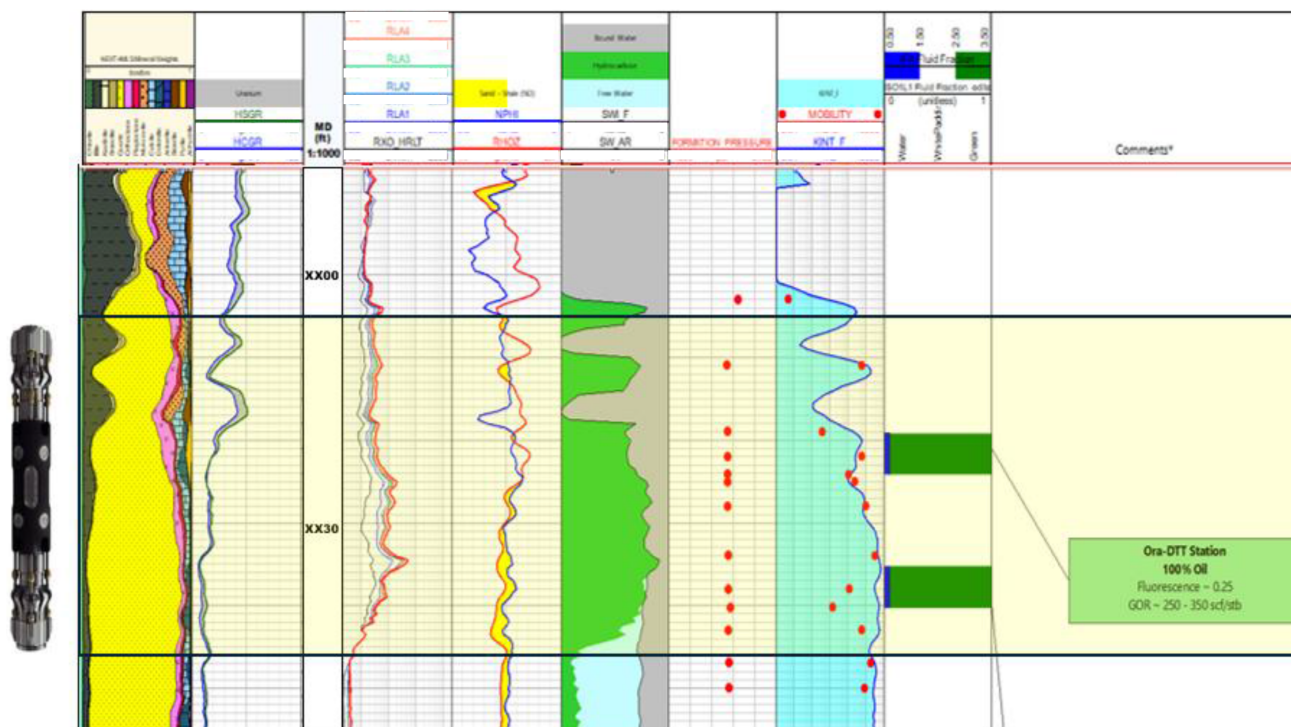


Figure 12—Focused radial probe set at XX24 ft – DTT Interval

The [table 1](#) that follows outlines the parameters and input values used in the analytical model. In the model, **wellbore storage was treated as a constant** coefficient because a down-hole shut-in valve present in the focused radial probe limited the compressible fluid volume and the flowing pressure remained above the bubble point, so no free gas accumulated in the flowline. A **limited-entry geometry** was selected: the focused radial probe contacts on a small portion (~2 ft) of the 40-ft net pay identified on open-hole logs, so the effective producing length is much shorter than the total interval. The reservoir interval was modeled as **homogeneous** because log derived porosity, permeability, and saturation show little vertical variability across the contributing sands. Finally, a **constant-pressure outer boundary model** was selected; the late-time derivative dips downward, a diagnostic that indicates a strong pressure support from an aquifer or large connected volume. The log indicates a water-bearing interval immediately below the oil column, which is interpreted as an active aquifer providing strong pressure support to the overlaying petroleum-bearing sands.

Table 1—Parameters used for analytical model

Data	Value	Data Source	Analysis Tools	
Average Porosity, ϕ	37%	OH logs	Wellbore Storage	Constant
Net thickness, h	~40 ft	OH Logs	Well Model	Limited Entry
Viscosity, μ	5 cP	Downhole Fluid Analyzer (FISO)	Reservoir model	Homogeneous
Total Compressibility, C_t	7.8×10^{-6}	PVT Correlations	Boundary model	Strong constant pressure
FCRF flow interval	Constant	Tool Schematic		
Well deviation at zone of interest, θ	0 deg	Vertical well		
Well Radius	4.25 inch	Well Schematic		

History Match of Data Using Analytical Model

The raw pressure data observed as in [Fig. 13](#) was clean with minimal noise. The analytical model demonstrated an excellent fit to the raw data across the early, middle and later time periods. The pressure response initially exhibited long spherical flow, transitioning into a brief radial-flow regime – the critical interval for estimating permeability. History matching between pressure and flow rate was satisfactory, particularly at late times when permeability estimates are most reliable. Early drawdowns appeared affected by near-wellbore skin, as the modeled drawdowns were lower than observed as in [Fig 14](#); however, this mismatch diminished as testing progressed, indicating the model's improved accuracy during extended drawdown and buildup phases.

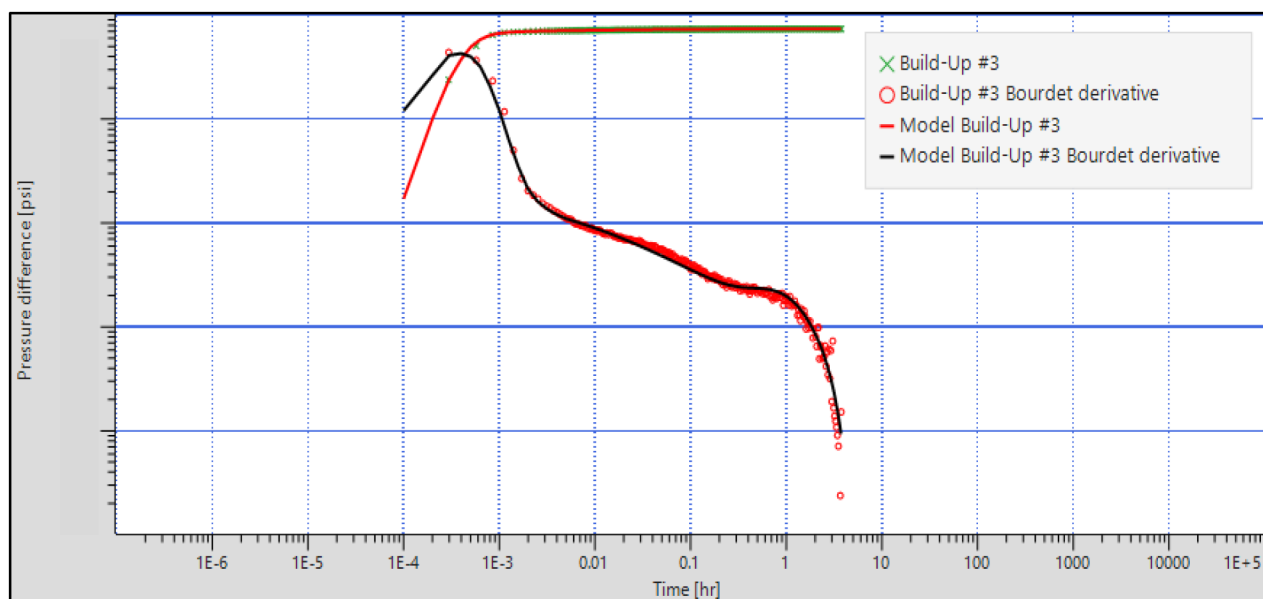


Figure 13—Analytical model – Pressure Transient Analysis

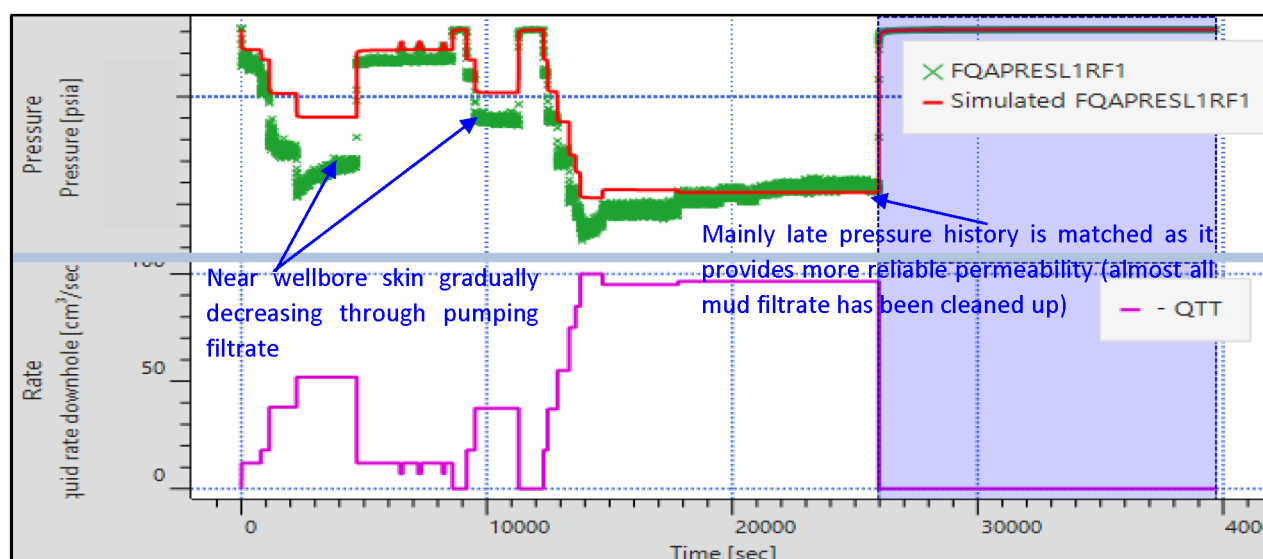


Figure 14—Actual & modelled pressure vs rate data

Semilog Horner Plot

A single regression line was applied across the radial-flow window on a Horner semi-log plot to estimate reservoir permeability. This method is reliable because during radial flow, pressure data tend to align linearly when plotted on a Horner time scale. The slope of this straight line reflects the reservoir's transmissibility, from which permeability can be calculated using standard Horner analysis equations. This technique is particularly effective when the radial flow period is short but clearly distinguishable, as shown in the provided Fig 15.

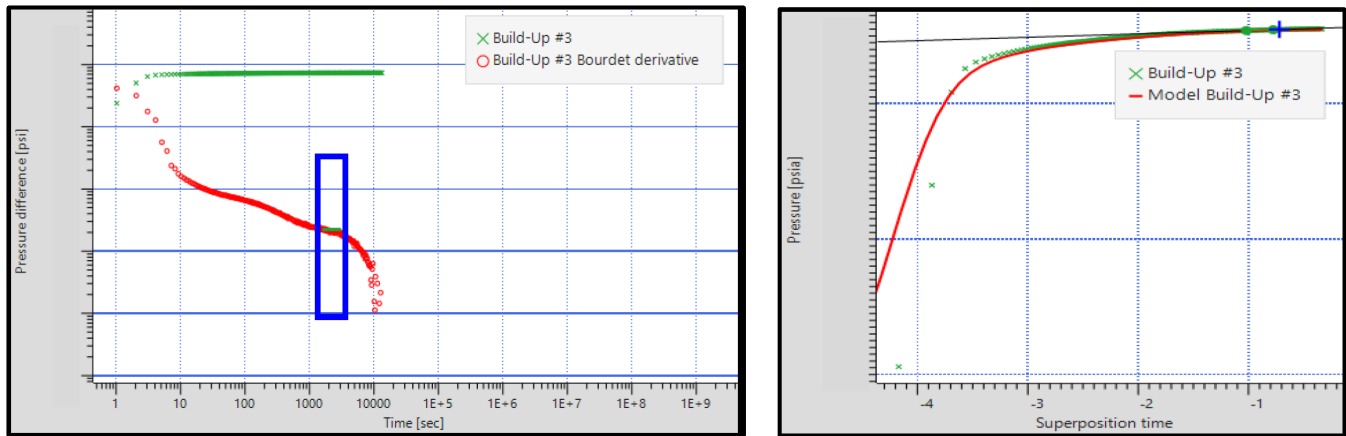


Figure 15—Horner Plot to estimate permeability

$$k = \frac{162.6 \mu B q}{m h}$$

Where k = permeability (mD), μ = viscosity (cP), B = formation volume factor (RB/STB), q = flowrate (STB/day), m = slope of horner plot (psi per log cycle) and h – net pay thickness (ft)

Radius of Investigation

The concept of radius of investigation (r_i) has been used in well testing for a long time (Mustapha *et al.*, 2023). The detailed theory behind the radius of investigation (r_i) is summarized in Kuchuk (2009). Throughout the oil and gas industry, the van Poolen (1964) equation is the most used method to evaluate r_i which is stated as below (Eqn 1):

$$r_i = 0.029 \sqrt{\frac{kt}{\phi \mu C_t}} \quad (\text{Eqn 1})$$

This equation only accounts for permeability (k), porosity (Φ), fluid viscosity (μ), and total compressibility (C_t). **The equation does not consider flow rate (q), gauge resolution (Δp), and noise and assumes the reservoir has a circular closed boundary.** The distance to the boundary is estimated where the pseudo steady state appears on the derivative plot.

Kuchuk (2009) proposed an equation (Eqn 2) shown below that accounts for flow rate and apparent gauge resolution. The approach from Kuchuk is to determine the furthest distance that the pressure change ΔP caused by the constant drawdown at the wellbore could be detected inside the reservoir. Kuchuk's equation shows the diffusion distance where the pressure change at the edge is bigger or equal to ΔP

$$r_i = 0.0459 \sqrt{\frac{kt}{\phi \mu C_t}} \sqrt{0.5 \ln\left(\frac{qt}{\Delta p h \phi C_t}\right) - \ln(r_i) - 1.2987} \quad (\text{Eqn 2})$$

Baед on the Kuchuk, van Poolen r_i gives a conservative value about 20% to 50% lower than the simulation-based approach. This method shows the importance of flowrate, flow duration, buildup duration, and fault type. There are several ways by which r_i can be increased. Increasing the flow rate and flow duration will increase the signal magnitude, increasing the signal-to-noise ratio and reducing the noise envelope. Similarly, decreasing the noise will also increase signal-to-noise ratio. The new formation tester system includes high-resolution gauges that reduce pressure noise.

The above Eqn (2) was used to calculate radius of investigation. Permeability was obtained from the results of analytical model and all the other parameters – porosity, compressibility, viscosity were taken from table 1 as shown above.

Deep Transient Testing (DTT) Summary & Results

Deep Transient Testing (DTT) was able to reach till around 1000 ft inside the reservoir without observing any reservoir boundaries. DTT is revealing high permeability rock (Darcy rock) that is laterally extended showing good potential for further development. Towards the end of the pressure derivative, pressure support is observed as the derivative tends to fall down. This is probably due to the water aquifer observed from the open hole logs.

RESULTS	
Initial Pressure (Pi)*	XX31 psi (reservoir pressure was within the expected range for this depth interval)
Permeability (k) (from Analytical model) *	Darcy range permeability
Permeability (k) (from Semilog plot) *	Darcy range permeability
Radius of Investigation (from Kuchuk's method)	~ 1000 ft
Skin	0.3

* To protect proprietary information, specific operational parameters, results (e.g.: Initial Pressure & Permeability in the above table) and log scales in figures 7, 8, 9, 10, 12, 13, 14 & 15 are not disclosed.

Conclusion

- **Proven Effectiveness of DTT with New Generation Formation Tester, Bridging the IPTT - DST gap**

Conducting Deep Transient Testing (DTT) with new generation formation tester platform demonstrated exceptional reservoir scale insight – revealing Darcy range permeability and ~ 1000 ft laterally extended reservoir without observing any boundaries. This case study makes a solid foundation for DTT which effectively bridges the gap between conventional Interval Pressure Transient Tests (IPTT) and full-scale Drill Stem Tests (DST), offering data quality comparable to DST with reduced operational impact.

- **Efficient, Safe Operation with New Generation Formation Tester**

The integrated use of focused sampling, dual high-rate pumps, and embedded automation enabled clean, minimized stationary time drastically while consistently yielding 100 % oil samples with zero contamination. A circulating sub incorporated in the toolstring kept the well over-balanced and continuously blended produced hydrocarbons into the mud stream for controlled return to surface, further enhancing operational safety.

- **Collaborative Multidisciplinary Workflow**

This paper integrates geology, petrophysics, and reservoir engineering teams into a cohesive process – leveraging each discipline's insights to interpret logs, select optimum sampling points, and precisely position the final DTT interval.

- **High-Potential, multi-Darcy reservoir confirmed for follow-up exploration:** DTT data verified Darcy range permeability and demonstrated excellent lateral continuity and strong aquifer support. Fluid analysis indicated light, high-value crude (~ 24 – 28 API) and the flow performance yielded a

good productivity index. These combined characteristics – high permeability, premium oil quality and robust deliverability – establish the tested interval as an attractive candidate for additional exploration and appraisal drilling in surrounding area.

- **Strategic role of DTT in Field Development Planning:** Conducting DTT in an exploration well delivers reservoir scale parameters. These insights are critical for optimizing well placement, spacing, and completion design. By integrating DTT results into early field modeling, operators can confidently identify appraisal well locations and design a layout that maximizes reservoir recovery and minimizes uncertainty, directly impacting investment decisions.

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